

MIDTERM PROJECT PROPOSAL

SCHOLAR AGENT: AGENTIC PIPELINE FOR LITERATURE REVIEW

INTRODUCTION

METHODOLOGY

EXPECTED RESULT

SUBJECT : AI SYSTEM II (8836556-01)

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01

Introduction

Problem and Motivation

Challenges in manual literature review

- Researchers face recurring obstacles when searching and synthesizing academic papers – motivating the need for an AI agent pipeline [\[1\]](#) [\[2\]](#).

1. Volume overload

Too many papers are published daily for any researcher to read manually.

2. Keyword blind spots

Searching one term misses papers that use different vocabulary for the same concept.

3. Information heterogeneity

Key details like method, dataset, and metric are structured differently in every paper.

4. Citation chasing

Tracing references forward and backward across hundreds of papers is tedious and error-prone.

5. Long-tail filtering

Researchers must skim dozens of papers just to find the few that match their specific criteria.



Proposed Solution

- An AI agent pipeline queries academic APIs with auto-generated search terms to discover relevant papers at scale [\[3\]](#)
- A Search Agent retrieves and ranks the top-K papers by metadata, removing the need for manual filtering [\[4\]](#).
- A Summary Agent extracts key information from each paper into a consistent structured format [\[5\]](#).
- An LLM as a Judge verifies that each output is accurate and faithful to the original paper before finalizing the output [\[6\]](#).

02

Research Methodology

Proposed Method and Expected Result

Proposed Method

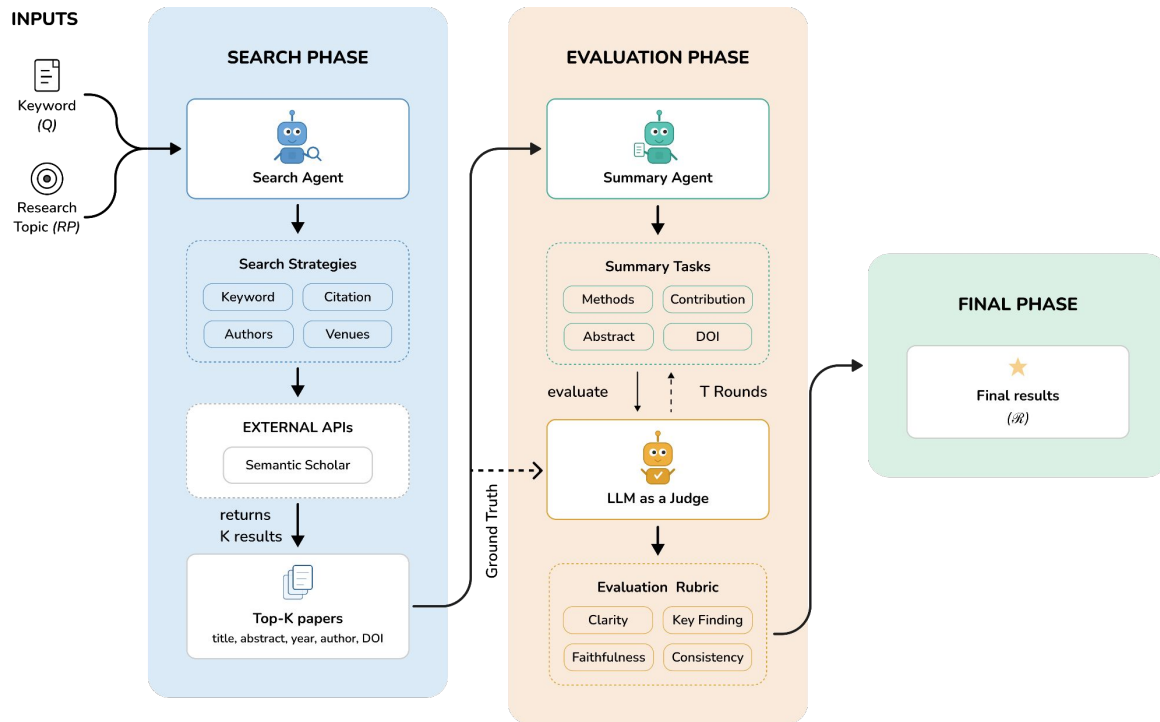


Figure 1. Scholar Agent System Architecture Diagram.

Expected Result

Table 1. Result of Papers' Information Conducted by Agents.

Title	Abstract	Published Year	DOI	Methodology	Contribution
Fill by Agent	Fill by Agent	Fill by Agent	Fill by Agent	Fill by Agent	Fill by Agent
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Our Limitations

- There is no ground truth available to objectively verify whether the retrieved papers are truly the most relevant for a given query.
- The LLM Judge cannot guarantee factual accuracy of summaries, as it relies solely on the model's own understanding of the paper.
- The pipeline depends on a single language model for both summarization and evaluation, which may introduce some bias or blind spots across all outputs.

03

References

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Thank You!

